Articulate3D Challenge — Technical Report

Frozen USDNet, Geometric Priors, and Structural Post-Processing

Junyuan Fang

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Team

Name	Affiliation	Email
Junyuan Fang	Independent Researcher	`fangjunyuan1@gmail.com`

Final standing (boards frozen 2026-08-28): **1st** on Movable-dev ($AP_{50}^{axis+origin} = 0.5038$) and Interaction-dev ($AP_{50} = 0.6446$); **2nd** on Movable-test (0.4056) and Interaction-test (0.3286). The extended report, "Geometric Refinement and Structural Post-Processing for Scene-Level Articulated Part Understanding," is publicly available in the code repository.

Method

The initial recipe (Fig. 1) freezes the official USDNet segmentation pathway; later full-network finetuning is a separate stage. The movable ranking metric is joint AP under mask, axis and origin matching constraints. Per-instance error analysis indicated different failure patterns: the regressed *axis* collapses onto the dominant gravity-aligned mode, and the regressed *origin* shrinks toward the part centroid.

1. **Initial frozen segmentation.** Minkowski U-Net backbone, Mask3D decoder, mask head and class head are kept at the released weights, *including BatchNorm running statistics*. AP_{50} is therefore bit-identical to the baseline within a checkpoint for articulation-only edits. Structural splitting and later full-network finetuning explicitly change the mask predictions.
2. **One trained component.** An 88 K-parameter attention-pooled origin head is finetuned with everything else frozen. Two silent pitfalls we hit and fixed: (a) frozen parameters do not stop BatchNorm buffers from drifting, which fakes gains on validation; (b) resuming a released checkpoint restarts USDNet's coarse-to-fine mask curriculum, which dilates the GT masks by 0.1 m for 100 epochs and destroys the model unless disabled.
3. **Training-free geometry overwrites the regressors.** Translation axis = PCA panel normal of the predicted mask, or in-plane if a sliding partner leaf exists; rotation axis forced vertical; rotation origin = mask endpoint perpendicular to the axis, on the side opposite the predicted handle (handles come from the companion interactable-part model). The initial head-plus-geometry recipe takes test $AP_{50}^{axis+origin}$ from 0.264 to 0.3385 with the segmentation branch untouched.
4. **Structural post-processing (both tasks).** The model routinely merges the two leaves of a double door into one instance. Offline DBSCAN connected-component splitting + fragment filter + NMS, tuned with the official evaluator on validation, improves test scores (movable $0.3385 \rightarrow 0.3760$; interaction $0.2686 \rightarrow 0.2886$).
5. **Full-network finetuning and ensemble.** Later train+validation finetunes update the full network with flip augmentation and the curriculum disabled. Interaction: the `v5`+`v6` union reaches 0.3240; adding `v5b` gives 0.3286. Movable: a single `mov_v11` checkpoint with geometric and structural refinement gives 0.4056. Validation is in-sample; these cumulative competition gains are not controlled multi-seed ablations.

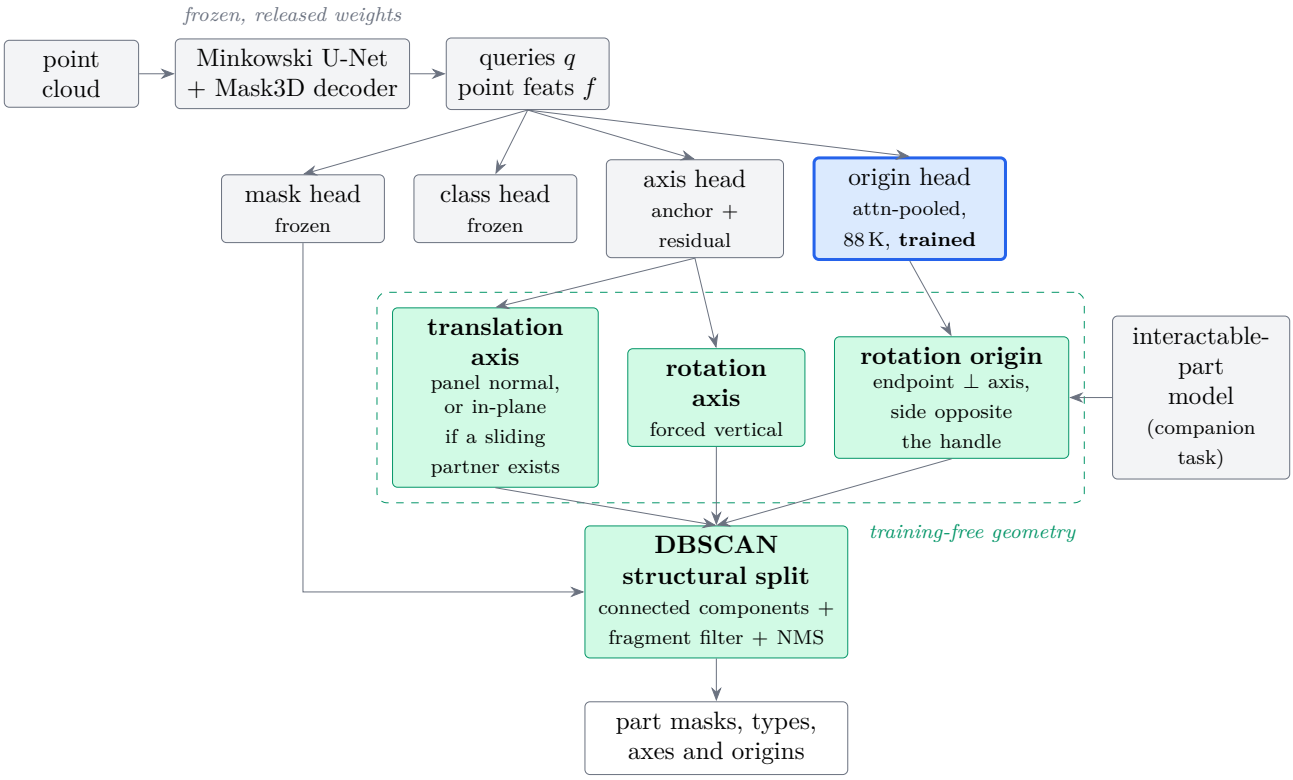


Figure 1: **Method overview.** Initial frozen-model stage only. Grey: frozen at released weights. Blue: the trained origin component. Green: training-free geometry read off the predicted masks and the offline structural post-processing.

Results (challenge server, ranking metric)

Configuration (cumulative)	Movable test $AP_{50}^{axis+origin}$	Interaction test AP_{50}
USDNet baseline (reproduced)	0.2640	0.2686
+ trained origin head + geometric priors	0.3385	—
+ DBSCAN structural post-processing	0.3760	0.2886
+ flip finetune / ensemble (final)	**0.4056**	**0.3286**
Relative to baseline	+54%	+22%